MAGNET-IoT: An Edge-Ready Graph Belief Extraction Approach for Lightweight Explainable IoT Anomaly Detection

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***Abstract*—The rapid proliferation of Internet of Things (IoT) devices has greatly increased the attack surface of modern networked systems, making real-time anomaly detection both critical and difficult. Most existing methods are supervised and require extensive labeled attack traffic, or they rely on computationally expensive deep learning and graph neural network (GNN) architectures that are unsuited for resource-constrained edge hardware. Many also operate as black boxes, preventing operators from understanding the evidence behind an alarm. We present MAGNET-IoT (Multi-scale Anomaly detection via Graph-belief Network for IoT), a normal-only-trained, graph-based anomaly detection framework with validation-based threshold calibration, designed for lightweight, explainable, and edge-deployable IoT security. MAGNET-IoT converts each five-second traffic window into a weighted communication graph and extracts ten interpretable *graph beliefs* that encode structural, behavioral, and information-theoretic network patterns. Rolling means at three time horizons provide multi-scale temporal context without recurrent networks, and instantaneous, temporal, and interaction features filtered by variance stability form a compact representation. The anomaly scorer is an Isolation Forest trained exclusively on normal traffic. On the Bot-IoT dataset, across five random seeds, MAGNET-IoT achieves a mean F1-score of 0.9521 ± 0.0124, AUROC of 0.9447 ± 0.0079, and balanced accuracy of 0.8686 ± 0.0109, with MCC of 0.3003 ± 0.0394 and Cohen's kappa of 0.1980 ± 0.0469. The best-performing seed (seed 42) reaches F1 $= 0.9696$, AUROC $= 0.9548$, MCC $= 0.3568$, and Cohen's kappa $= 0.2675$. On the independent TON-IoT dataset, using dataset-specific normal-only training and validation-based threshold calibration, it achieves an F1-score of 0.9867, recall of 1.0000, and AUROC of 0.9942, demonstrating the applicability of the pipeline to a second IoT dataset. The model occupies 1.63 MB with an inference latency of 0.043 ms per window on commodity CPU hardware. Compared with modern temporal and graph representation-learning baselines (LSTM, TCN, and Graph Autoencoders), MAGNET-IoT achieves a superior balance of accuracy, latency, and intrinsic explainability: the LSTM and TCN autoencoders underperform in recall and AUROC, while the polarity-corrected GAE reaches a comparable F1 but incurs $57\times$ the inference latency, weaker AUROC, and no interpretability. Global explainability is provided through permutation-based belief importance, revealing that *unique_dst_ratio* (27.1%), *flow_burstiness* (16.5%), and *protocol_entropy* (13.6%) are the dominant anomaly indicators. Ablation, correlation, and hyperparameter sensitivity studies validate the design. MAGNET-IoT provides a practical, explainable, and computationally efficient foundation for reliable IoT security at the network edge.**

***Index Terms*—IoT anomaly detection, graph-based feature extraction, unsupervised learning, edge computing, explainability, Isolation Forest, graph belief extraction, lightweight security, Bot-IoT, TON-IoT, temporal modeling.**

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Fig. 1. Application domains of IoT anomaly detection.

I. Introduction

THE Internet of Things (IoT) is one of the most important technological ideas of the 21st century [1]. Billions of different devices now talk to each other over shared networks, creating constant streams of machine-to-machine traffic. These devices range from smart homes and industrial control systems to precision agriculture and connected healthcare. Recent reports from the industry say that by 2030, there will be more than 30 billion active IoT devices, and the amount of data they send will grow by more than 28% each year. This unprecedented scale brings about transformative benefits like real-time monitoring, autonomous decision-making, and predictive maintenance. However, it also creates a huge attack surface that enemies are increasingly taking advantage of. IoT deployments are different from traditional enterprise networks in that they often involve devices with limited resources, different communication protocols, and long periods of unattended operation. This makes it harder to use traditional security measures. Because of this, a single compromised sensor or gateway can be used to move laterally, steal data, or recruit a large number of bots, so strong, real-time anomaly detection is not just an improvement, but a must-have for reliable IoT ecosystems.

Fig. 1 illustrates the wide range of areas where IoT-based anomaly detection is directly important [2]. In recent years, there has been a significant rise in IoT-based cyberattacks. In 2025, the FBI warned about BADBOX 2.0, a botnet that affected millions of home IoT devices to launch proxy services and DDoS attacks [3]. Similarly, the "Matrix" botnet took advantage of vulnerable IP cameras and routers to form a global DDoS-for-hire network [4]. Even more alarming, the state-sponsored "Raptor Train" botnet has been operating since 2020 and has compromised over 200,000 devices using a customized version of Mirai [5]. These attacks share common traits: they exploit weak device security, unpatched vulnerabilities (e.g., AVTECH IP cameras [6]), and show up as abnormal network behaviors such as scanning, changes in traffic patterns, and unusual data volumes. The vast amount of IoT data—shown by the 2.7 billion records exposed in the Mars Hydro breach [1]—highlights the urgent need for simple, real-time, and understandable anomaly detection systems.

This diversity brings serious security challenges. IoT devices are often compromised in Distributed Denial-of-Service (DDoS) amplification campaigns, botnet formation, and data theft attacks [7]. However, the typical security tools do not perform well in this area. Four major challenges stand out.

Scarcity of labeled data. Supervised intrusion detection systems need large, representative sets of labeled attack traffic. In real-world IoT setups, such labels are often hard to find at scale: normal behavior is abundant, but confirmed attack examples are sparse, delayed, or entirely absent [8]. Semi-supervised and few-shot approaches mitigate but do not eliminate this dependency [9].

Resource constraints. Modern GNN-based detectors achieve high accuracy but require a lot of GPU memory, multi-hour training cycles, and model sizes in the hundreds of megabytes [10]–[12]. Most IoT edge nodes—microcontrollers, single-board computers, embedded gateways—cannot host such models. Lightweight alternatives exist [13], [14] but sacrifice temporal reasoning or graph-structural insight.

Lack of explainability. Black-box detectors raise concerns in operational contexts: a security analyst cannot act on an alarm without understanding *which* traffic features triggered it. Post-hoc explainers such as SHAP and LIME add computational overhead and introduce approximation error [15], [16]. A framework with intrinsic, model-native explanations is therefore preferable [17], [18].

Neglect of temporal patterns. Network attacks occur over time. Scanning, DDoS ramp-up, and command-and-control beaconing all exhibit temporal signatures invisible to per-packet or static-window methods. Recurrent architectures capture these dynamics but add training complexity and latency [19], [20].

MAGNET-IoT is designed to solve these four interconnected challenges in one framework. We now explicitly map each design choice to the challenge it addresses:

1) **Label scarcity:** MAGNET-IoT trains its Isolation Forest exclusively on normal traffic (367 normal windows on Bot-IoT and 59 normal windows on TON-IoT). No attack labels are used during Isolation Forest training; labels are used only for threshold calibration on the validation set and for test evaluation. *Evidence:* across five random seeds on Bot-IoT, mean F1 $= 0.9521 \pm 0.0124$; best seed F1 $= 0.9696$; TON-IoT F1 $= 0.9867$.
2) **Resource constraints:** Instead of recurrent or graph neural networks, MAGNET-IoT uses interpretable graph beliefs, rolling means, and an Isolation Forest. The model is 1.63 MB with 0.043 ms inference latency on CPU. *Evidence:* Table VIII and Fig. 13.
3) **Lack of explainability:** MAGNET-IoT provides intrinsic global and local explanations through permutation importance and z-score deviations, without external XAI libraries. *Evidence:* Figs. 9, 10, and 11.
4) **Neglect of temporal patterns:** Multi-scale rolling means over 5 s, 15 s, and 30 s capture temporal context without recurrent-network inference overhead. *Evidence:* the temporal ablation and comparison against LSTM/TCN in Table IV.

The main contributions of this study are:

1) A graph belief representation framework that organizes ten interpretable structural, behavioral, and information-theoretic features into a compact anomaly representation optimized for edge IoT environments.
2) A lightweight temporal feature engineering scheme using multi-scale rolling means, avoiding the computational overhead and overfitting risks of recurrent networks.
3) A normal-only-trained detection pipeline, validated on two datasets (Bot-IoT and TON-IoT) using dataset-specific normal-only training and validation-based threshold calibration.
4) Intrinsic explainability via permutation importance and local z-score deviations, providing actionable evidence without external XAI tools.
5) A configurable threshold mechanism supporting high-recall, balanced, and intermediate operating modes.
6) Extensive evaluation against modern graph and temporal baselines (GAE, LSTM, TCN), traditional anomaly detectors, and raw-feature baselines, including ablation, correlation, hyperparameter, and five-seed statistical analyses.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents MAGNET-IoT. Section IV describes the experimental setup. Section V reports and analyzes results. Section VI discusses implications and limitations. Section VII concludes the paper.

II. Related Work

A systematic literature review by Boutassetta et al. [21] categorises AI-based cyberattack detection for smart cities into ML-based, DL-based, and hybrid approaches, covering over 40 papers. The review identifies critical gaps: most methods lack real-world validation, edge deployment remains challenging due to high computational demands, and simpler ML approaches often ignore temporal modelling. This reinforces the need for lightweight, temporal, edge-deployable solutions such as the one proposed in this work.

A. Graph-Based IoT Anomaly Detection

Graph neural networks are now the most popular way to find network-level anomalies because they can model relational structure. Rivera and Uribe [11] propose a spatio-temporal GNN combining GCN, GAT, and LSTM for IoT anomaly detection, achieving 96.3% accuracy on Bot-IoT and TON-IoT. However, their framework is semi-supervised (requiring 1–5% labeled data), demands GPU training, and provides no explanation of individual detections. Villegas-Ch et al. [22] train a three-layer GCN on a private simulated dataset of 10 million records using a Tesla V100, reporting F1 $= 0.95$ for normal traffic—but the non-public dataset makes results non-reproducible, and the method produces no interpretable output. George and Thampi [2] represent devices as nodes and known CVE exploits as weighted edges, enabling offline attack-path identification in industrial IoT. While elegant, the approach is static, depends on known vulnerability databases, and does not address real-time traffic anomaly detection. Ahanger et al. [23] propose a GAT-based IDS on NSL-KDD data, but the dataset is outdated (2009) and not IoT-specific.

MAGE [10] advances the state of the art with FFT-based multiperiodic segmentation, dynamic graph structure learning, and a Transformer discrepancy module, achieving F1 $= 0.9777$ on SWaT. FuGLAD [12] fuses three complementary graph structure learning objectives with a DCRNN predictor, outperforming 13 baselines across five benchmark datasets. Both represent powerful contributions, but neither offers explainability, and both require GPU resources that preclude edge deployment. Nguyen et al. [24] propose PSI-Graph for static IoT botnet analysis across CPU architectures, but the method is supervised and post-mortem only.

B. Unsupervised and Lightweight Detection

Guo et al. [25] combine a variational autoencoder, GRU, and Gaussian mixture model for unsupervised IoT anomaly detection, reporting F1 up to 0.9567 on the Intel dataset. The model is GPU-dependent and offers no interpretability. Christopher et al. [8] address class imbalance via EG-SMOTE and propose GSOM-HD for edge-deployed unsupervised detection (F1 ≈ 0.945 on Bot-IoT), but provide only binary output with no explanation. Shao et al. [14] achieve low-latency detection using Bloom filters and FJLT-based locality-sensitive hashing, outperforming Isolation Forest in throughput, but the method is semi-supervised and lacks temporal or graph-structural modeling. Ma et al. [13] demonstrate a Raspberry Pi-deployable botnet detector with 52% latency reduction using CDF + Gini feature selection, confirming feasibility of truly lightweight IoT security—though without explainability. Alam et al. [26] propose a federated transfer learning framework for smart city threat intelligence, but provide no experimental results or performance metrics. Houichi et al. [27] propose a federated learning framework that achieves only 68.5% accuracy and detects zero attacks due to unaddressed class imbalance. Sarker Emon et al. [28] present CityShield, a theoretical multi-layered threat detection architecture; however, the work remains conceptual, as it lacks implementation details and experimental validation.

C. Explainability in IoT Security

The need for interpretable IoT intrusion detection is well documented. Moustafa et al. [7] survey the literature and identify that 69.2% of ML-based IDS papers use outdated datasets, existing XAI methods assume feature independence, and real-time explainability at the edge remains an open challenge. Muna et al. [15] apply SHAP and LIME to XGBoost predictions on IOTD20, providing feature-level post-hoc explanations—but at computational overhead incompatible with edge deployment. Oseni et al. [16] combine CNN with Deep SHAP for 99.15% accuracy on TON-IoT, but the method is supervised, GPU-dependent, and relies on an external XAI library rather than intrinsic model transparency. Costa et al. [17] propose SECURER, a rule-based supervised classifier for device behavior that is inherently explainable and robust to missing data, but limited to device on/off patterns and unable to analyze network-level attacks. Ambritta et al. [18] survey XAI selection strategies for ethical IoT systems, confirming that intrinsic, lightweight explainability remains the foremost unsolved challenge. Li et al. [29] propose model-agnostic global rule extraction via IC-Tree decomposition, but the method adds a post-hoc computational layer.

D. Temporal Modeling Without Recurrent Networks

Wang et al. [19] represent one of the closest supervised baselines: an RNN/LSTM network optimized with Particle Swarm Optimization for real-time smart-city threat detection, achieving 95.7% accuracy and 94.2 ms latency. Despite temporal modeling capability, the method is supervised, requires 5.2 hours of GPU training, uses an unspecified UCI dataset, and provides no explanation. Wang et al. [20] combine Dueling DQN and VAE for weakly supervised traffic anomaly detection, improving AUC-PR by 12.2% over baselines on UNSW-NB15—but remain GPU-dependent and lack both temporal graph structure and explainability. Almaraz-Rivera et al. [30] use self-supervised MoCo v2 on IoT flows converted to images, but still require hundreds of labeled samples and GPU training.

E. Positioning of MAGNET-IoT

Table I summarizes the key dimensions across selected prior works. As the table shows, no existing method simultaneously satisfies all four desiderata: unsupervised operation, temporal modeling without recurrent networks, lightweight edge deployment (< 2 MB, < 0.1 ms), and native explainability without external XAI libraries. Specifically, graph-based methods such as [10] and [11] achieve strong detection but require GPU resources and offer no explainability. Lightweight alternatives such as [13] sacrifice temporal reasoning and unsupervised operation. Explainable methods such as [17] and [29] remain either supervised or lack temporal and graph-structural modeling. This gap motivates the design of MAGNET-IoT, presented in Section III, which is evaluated against these dimensions in Section V.

TABLE I
LIMITATIONS OF EXISTING METHODS ACROSS KEY DIMENSIONS

Method	Unsup.	Temporal	Lightweight	Explainable
[10]	✓	✓	×	×
[11]	×	✓	×	×
[13]	×	×	✓	×
[17]	×	×	✓	✓
[19]	×	✓	×	×
[25]	✓	✓	×	×
[29]	✓	×	✓	✓

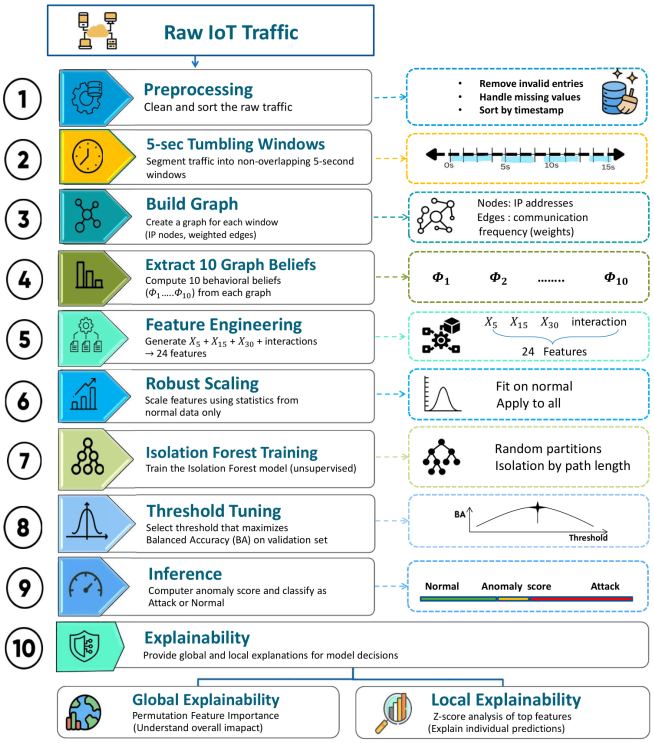


Fig. 2. MAGNET-IoT pipeline overview.

III. METHODOLOGY

Fig. 2 provides an overview of the MAGNET-IoT pipeline. The framework comprises six sequential stages: data ingestion and preprocessing, temporal windowing, graph construction and belief extraction, multi-scale feature engineering, unsupervised model training, and threshold-optimized inference with explainability output.

A. Data Ingestion and Preprocessing

Let the raw dataset consist of N network flow records $\mathcal{F} = \{f_1, f_2, \ldots, f_N\}$, where each record denoted $f_i = (t_i, \text{src}_i, \text{dst}_i, \pi_i, b_i, y_i)$ contains timestamp t_i, source IP src_i, destination IP dst_i, protocol π_i, byte count b_i, and binary label $y_i \in \{0, 1\}$. Labels are only used for threshold tuning and evaluation; they do not go into the Isolation Forest training process.

Preprocessing steps include (i) numeric type coercion for timestamps and byte counts, (ii) removal of records with missing or malformed source/destination addresses, (iii) deduplication of exact timestamp-source-destination triples, (iv) putting IP and protocol fields in alphabetical order to save memory, and (v) sorting by time to keep the flow order for downstream windowing.

B. Temporal Windowing

Flows are partitioned into non-overlapping *tumbling windows* of fixed duration $\Delta = 5$ seconds. For a dataset spanning $[t_{\min}, t_{\max}]$, window w_k collects all flows with

$$t_{\min} + k\Delta \;\le\; t_i \;<\; t_{\min} + (k+1)\Delta, \quad k = 0, 1, 2, \ldots \tag{1}$$

Each window is labeled as anomalous ($y_{w_k} = 1$) if it contains at least one attack flow, and normal ($y_{w_k} = 0$) otherwise. This window-level labeling is in line with operational detection, where the response level is a traffic epoch instead of a single packet.

C. Graph Construction

For each window w_k, a weighted undirected graph $G_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{W}_k)$ is constructed as follows. The node set $\mathcal{V}_k$ is the union of all source and destination IP addresses observed in w_k. An edge $(u, v) \in \mathcal{E}_k$ exists if at least one flow from u to v (or v to u) occurred in the window, and its weight W_{uv} equals the total number of flows between u and v:

$$W_{uv} = \sum_{f_i \in w_k} \mathbf{1}[\{\text{src}_i, \text{dst}_i\} = \{u, v\}]. \tag{2}$$

D. Graph Belief Extraction

Ten scalar *graph beliefs* $\boldsymbol{\phi}_k \in \mathbb{R}^{10}$ are computed from G_k. These beliefs are designed to be interpretable, computationally inexpensive, and behaviorally discriminative.

1) Protocol Entropy (ϕ_1):

$$\phi_1 = -\sum_{p} \frac{n_p}{|\mathcal{F}_k|} \log_2 \frac{n_p}{|\mathcal{F}_k|} \tag{3}$$

where n_p is the count of flows using protocol p in window w_k. High entropy indicates protocol diversity characteristic of scanning attacks where attackers probe multiple service ports across TCP, UDP, and ICMP.

2) Edge Entropy (ϕ_2):

$$\phi_2 = -\sum_{(u,v)\in\mathcal{E}_k} \frac{W_{uv}}{W_{\text{tot}}} \log_2 \frac{W_{uv}}{W_{\text{tot}}}, \quad W_{\text{tot}} = \sum_{(u,v)} W_{uv} \tag{4}$$

Uniform edge entropy indicates distributed traffic; low entropy reveals concentration characteristic of targeted flooding where a single connection carries most flows.

3) Edge Density (ϕ_3):

$$\phi_3 = \frac{2|\mathcal{E}_k|}{|\mathcal{V}_k|(|\mathcal{V}_k| - 1)} \tag{5}$$

High density during brief windows signals abnormal communication fan-out, typical of worm propagation or DDoS reflection attacks.

4) Node Concentration (ϕ_4):

$$\phi_4 = \frac{\max_{v \in \mathcal{V}_k} \deg(v)}{|\mathcal{E}_k|} \tag{6}$$

A single hub dominating the graph topology is a strong indicator of command-and-control or amplification behavior in botnet operations.

5) Flow Burstiness (ϕ_5):

$$\phi_5 = \frac{W_{\max} - \bar{W}}{W_{\max} + \bar{W}}, \quad W_{\max} = \max_{(u,v)} W_{uv},\ \bar{W} = \frac{1}{|\mathcal{E}_k|} \sum_{(u,v)} W_{uv} \tag{7}$$

Values near 1 indicate that a single edge dominates all traffic, characteristic of UDP flooding or TCP SYN flood attacks.

6) Degree Skewness (ϕ_6):

$$\phi_6 = \frac{\frac{1}{|\mathcal{V}_k|} \sum_v (\deg(v) - \bar{d})^3}{\left(\frac{1}{|\mathcal{V}_k|} \sum_v (\deg(v) - \bar{d})^2 \right)^{3/2}} \tag{8}$$

Positive skew indicates star-like topologies where few high-degree nodes drive most traffic—a signature of DDoS reflectors and scanning proxies.

7) Average Clustering Coefficient (ϕ_7):

$$\phi_7 = \frac{1}{|\mathcal{V}_k|} \sum_{v \in \mathcal{V}_k} C(v), \quad C(v) = \frac{2\,|\{(u,w) \in \mathcal{E}_k : u, w \in \mathcal{N}(v)\}|}{\deg(v)(\deg(v) - 1)} \tag{9}$$

Benign device clusters maintain moderate clustering; attacks disrupt locality by creating star-shaped communication patterns.

8) Top-5 Traffic Ratio (ϕ_8):

$$\phi_8 = \frac{\sum_{v \in \mathcal{T}_5} s(v)}{W_{\text{tot}}}, \quad s(v) = \sum_{(v,u) \in \mathcal{E}_k} W_{vu} \tag{10}$$

where $\mathcal{T}_5$ is the set of five nodes with the highest total traffic $s(v)$. Traffic concentration in a small node subset indicates flooding attacks targeting specific IPs.

9) Mean Inter-Arrival Time (ϕ_9):

$$\phi_9 = \frac{1}{|\mathcal{F}_k| - 1} \sum_{i=1}^{|\mathcal{F}_k|-1} (t_{i+1} - t_i) \tag{11}$$

Abnormally short IATs indicate high-rate scanning or flooding, while abnormally long IATs may indicate slow C&C beaconing.

10) Unique Destination Ratio (ϕ_{10}):

$$\phi_{10} = \frac{|\{\text{dst}_i : f_i \in w_k\}|}{|\mathcal{F}_k|} \tag{12}$$

A near-unity ratio—many distinct destinations per flow—is the primary fingerprint of horizontal port scanning and DDoS enumeration. This belief emerged as the single most important feature in our analysis.

TABLE II
TAXONOMY OF GRAPH BELIEFS AND THEIR BEHAVIORAL RATIONALE

Category	Beliefs	Attack Behavior Captured
Structural	Edge Density, Average Clustering, Degree Skewness, Node Concentration	Abnormal topology. DDoS reflection and worm propagation create star-shaped graphs with high degree skewness and low clustering.
Behavioral	Flow Burstiness, Top-5 Ratio, Unique Destination Ratio	Traffic dynamics. Horizontal port scanning creates a near-unity unique destination ratio, while UDP flooding creates extreme flow burstiness.
Information	Protocol Entropy, Edge Entropy, Mean Inter-Arrival Time	Protocol/service diversity. Reconnaissance tools probing multiple ports increase protocol entropy, while slow C&C beaconing alters IAT means.

11) Belief Taxonomy and Design Rationale: To avoid the impression that the ten beliefs are an arbitrary collection of standard graph metrics, we organize them into three complementary categories. Table II summarizes the taxonomy and the attack behavior each category is intended to capture.

The taxonomy ensures that MAGNET-IoT captures structural, behavioral, and information-theoretic properties rather than redundant variants of the same graph statistic. The Spearman correlation analysis in Section V-H empirically confirms that most belief pairs are weakly to moderately correlated, although a few pairs exhibit strong correlations ($|\rho| > 0.8$); the ablation study in Section V-F nevertheless confirms that every belief category contributes measurably to detection performance.

E. Multi-Scale Temporal Feature Engineering

A single-window belief vector $\boldsymbol{\phi}_k$ captures the instantaneous network state but ignores temporal evolution. To model traffic dynamics across multiple horizons *without* recurrent networks, we compute rolling means at two additional scales:

$$\boldsymbol{\phi}_k^{(15)} = \frac{1}{\min(k+1, 3)} \sum_{j=\max(0,k-2)}^{k} \boldsymbol{\phi}_j \tag{13}$$

$$\boldsymbol{\phi}_k^{(30)} = \frac{1}{\min(k+1, 6)} \sum_{j=\max(0,k-5)}^{k} \boldsymbol{\phi}_j \tag{14}$$

corresponding to smoothed views over approximately 15 s and 30 s respectively. Stacking the three scales yields a 30-dimensional temporal feature block.

Seven cross-belief interaction features are then appended:

$$\begin{aligned} &\psi_1 = \phi_1 \cdot \phi_3, \quad \psi_2 = \phi_2 \cdot \phi_4, \\ &\psi_3 = \phi_3 \cdot \phi_5, \quad \psi_4 = \phi_6 \cdot \phi_8, \\ &\psi_5 = \phi_7 \cdot \phi_{10}, \quad \psi_6 = \phi_3/(\phi_4 + \epsilon), \quad \psi_7 = \phi_{10}/(\phi_5 + \epsilon) \end{aligned} \tag{15}$$

yielding a 37-dimensional raw feature vector. A variance filter (threshold $\sigma^2 > 10^{-6}$) fitted on training-normal windows removes zero-variance dimensions, yielding a dataset-dependent retained dimension d: $d = 24$ for Bot-IoT and $d = 34$ for TON-IoT (Section IV-A).

Algorithm 1 summarizes the complete feature engineering procedure.

Algorithm 1 MAGNET-IoT Feature Engineering

Require: Window sequence $\{w_0, w_1, \ldots, w_T\}$, belief extractor $\Phi(\cdot)$, variance mask $\mathbf{m}$ (fitted on training normals; dataset-specific)

Ensure: Feature matrix $\mathbf{X} \in \mathbb{R}^{T \times d}$, where d is the retained dimension (24 for Bot-IoT, 34 for TON-IoT)

for $k = 0$ **to** T **do**
  $\boldsymbol{\phi}_k \leftarrow \Phi(G_k)$ {Extract 10 graph beliefs}
  $\boldsymbol{\phi}_k^{(15)} \leftarrow \text{RollingMean}(\boldsymbol{\phi}, k, L{=}3)$
  $\boldsymbol{\phi}_k^{(30)} \leftarrow \text{RollingMean}(\boldsymbol{\phi}, k, L{=}6)$
  $\boldsymbol{\psi}_k \leftarrow \text{Interactions}(\boldsymbol{\phi}_k)$ {Eqs. (15)}
  $\tilde{\mathbf{x}}_k \leftarrow [\boldsymbol{\phi}_k,\ \boldsymbol{\phi}_k^{(15)},\ \boldsymbol{\phi}_k^{(30)},\ \boldsymbol{\psi}_k]$ {37-dim raw vector}
  $\mathbf{x}_k \leftarrow \tilde{\mathbf{x}}_k[\mathbf{m}]$ {Apply variance mask $\Rightarrow$ d-dim}
end for
return $\mathbf{X}$

F. Robust Scaling

The d-dimensional feature matrix is standardized using a RobustScaler fitted exclusively on training-normal windows. For each feature j:

$$\hat{x}_{kj} = \frac{x_{kj} - \text{median}_j}{Q_{75,j} - Q_{25,j}} \tag{16}$$

where $Q_{25,j}$ and $Q_{75,j}$ are the 25th and 75th percentiles computed over normal training windows. The interquartile range denominator provides robustness against the heavy-tailed distributions characteristic of network traffic. As with the variance mask, the scaler is fitted independently per dataset.

G. Unsupervised Anomaly Scoring with Isolation Forest

An Isolation Forest (iForest) [31] is trained on the scaled feature vectors of normal windows only. iForest builds an ensemble of random trees by recursively selecting a feature q uniformly at random and a split value v drawn uniformly from $[\min_q, \max_q]$; a point is isolated at depth proportional to $1/\log_2 n$, so anomalies—being sparse and different—are isolated early. The anomaly score for a test instance $\hat{\mathbf{x}}$ is:

$$s(\hat{\mathbf{x}}) = 2^{-\frac{E[h(\hat{\mathbf{x}})]}{c(n)}} \tag{17}$$

where $E[h(\hat{\mathbf{x}})]$ is the mean path length across trees and $c(n) = 2H(n-1) - (2(n-1)/n)$ is the average path length of unsuccessful BST searches for n samples. A higher negated decision function value indicates a stronger anomaly.

We use 200 trees, max_samples $= 256$, bootstrap sampling, and `contamination = auto`. These hyperparameters were fixed prior to test evaluation; the sensitivity analysis in Section V-G reports their diagnostic behavior on the test set and did not influence the configuration choice.

H. Threshold Optimization

A scalar threshold τ converts continuous anomaly scores to binary labels. Rather than applying a fixed percentile, we select τ by maximizing *balanced accuracy* (BA) on the held-out validation set:

$$\tau^* = \arg\max_{\tau}\ \text{BA}(\tau), \quad \text{BA} = \tfrac{1}{2}\left(\text{TPR} + \text{TNR}\right) \tag{18}$$

over a grid of 200 candidate values in $[0, \max_\tau s_{\text{val}}]$. This criterion explicitly penalizes both missed attacks and false alarms, making it well-suited to the severely imbalanced Bot-IoT distribution. The balanced threshold obtained in the reference run (seed 42) is $\tau^* = 0.1356$; each seed in the five-seed study tunes its own threshold.

I. Computational Complexity Analysis

For a window with V IP addresses (nodes) and E flows (edges), graph construction costs $O(E)$, and belief extraction costs $O(V + E)$ for degree, entropy, and ratio statistics; the average clustering coefficient costs $O(V \cdot \langle d \rangle^2)$ where $\langle d \rangle$ is the mean node degree (in our graphs, fewer than 50 nodes per window, so this cost is negligible in practice). Isolation Forest inference costs $O(T \log N)$, where $T = 200$ trees and $N = 256$ samples per tree. In contrast, a standard GNN layer requires $O(E \times d)$ operations, where d is the hidden dimension, plus GPU tensor overhead. This analysis explains why MAGNET-IoT remains lightweight even before empirical timing.

J. Explainability

1) Global Explainability — Permutation Importance: For each graph belief ϕ_j, permutation importance I_j is estimated by measuring the drop in balanced accuracy when the columns of $\mathbf{X}_{\text{val}}$ corresponding to belief j are randomly permuted:

$$I_j = \text{BA}(\mathbf{X}_{\text{val}}) - \mathbb{E}_\sigma\left[\text{BA}(\mathbf{X}_{\text{val}}^{(j,\sigma)})\right] \tag{19}$$

with 10 permutation repeats. Importances are clipped to zero and normalized to sum to 100%.

2) Local Explainability — Z-Score Deviation: For a test window w_k, each belief $\phi_{k,j}$ is compared to the empirical distribution of normal training windows:

$$z_{k,j} = \frac{\phi_{k,j} - \mu_j^{\text{norm}}}{\sigma_j^{\text{norm}}} \tag{20}$$

where μ_j^{norm} and σ_j^{norm} are the mean and standard deviation of belief j over all training-normal windows. The top-3 beliefs by $|z_{k,j}|$ are surfaced to the operator as the primary evidence for the anomaly decision, providing a human-readable, model-native explanation without any external XAI library.

IV. Experimental Setup

A. Datasets

Bot-IoT dataset. Experiments are conducted on the Bot-IoT dataset [32], a publicly available benchmark specifically designed for IoT intrusion detection. The dataset was created in a realistic testbed setting that has a wide range of normal IoT traffic and different types of attacks, like DDoS, DoS, reconnaissance, and data exfiltration. The raw data is spread out over four CSV files, containing a total of 4,000,000 flow records. A rigorous cleaning pipeline is applied to ensure data quality: deduplication of identical timestamp-source-destination triples, type coercion of timestamps and byte counts, and elimination of records with malformed or missing source/destination addresses. After cleaning, 3,406,910 valid

flows remain. The dataset exhibits severe class imbalance: only 7,136 flows (0.21%) represent normal traffic, while the remaining 99.79% correspond to various attack categories.

Following (1), flows are aggregated into non-overlapping 5-second tumbling windows, yielding 11,119 windows in total: 612 normal and 10,507 attack. To simulate a normal-only deployment scenario, we execute a stratified split at the window level: Training set: 367 normal windows (60% of all normal windows; no attack windows). Validation set: 122 normal windows + 2,101 attack windows, used exclusively for threshold tuning and model selection. Test set: 123 normal windows + 8,406 attack windows, held out for final evaluation. No attack labels are used during Isolation Forest training; labels are used only for threshold tuning and test evaluation.

TON-IoT dataset. To evaluate the applicability of the pipeline to a second IoT dataset, we evaluated MAGNET-IoT on the TON-IoT network dataset [33]. TON-IoT provides a modern testbed with a diverse attack taxonomy, including backdoor, DDoS, ransomware, and scanning traffic. Because the network-flow CSV evaluated in this study lacks a native timestamp, we derived a sequential index from row order and grouped consecutive flows into windows of 100 rows, yielding the same 5-second-equivalent grouping used on Bot-IoT. We caution that these are *row-order groups*, not verified wall-clock time windows; consequently, the multi-scale temporal features on TON-IoT reflect row-order groupings rather than true five-second intervals. This is a limitation of the available data and does not affect the Bot-IoT evaluation, where real timestamps are used. After cleaning, 211,043 flows formed 423 row-order groups (99 normal and 324 attack). These 423 groups were partitioned into 59 training groups (all normal), 84 validation groups (20 normal and 64 attack) used exclusively for threshold selection, and 280 test groups (20 normal and 260 attack). No attack labels are used during Isolation Forest training; labels are used only for threshold tuning and test evaluation. The variance mask and RobustScaler are fitted independently on the TON-IoT normal training groups, and the resulting retained dimension is $d = 34$ (vs. $d = 24$ on Bot-IoT), reflecting higher protocol and connection-state diversity.

B. Evaluation Metrics

Given the severe class imbalance in Bot-IoT (only 0.21% normal flows), standard accuracy is misleading. We report Precision, Recall (TPR), True Negative Rate (TNR), F1-score, Balanced Accuracy (BA), AUROC, Matthews Correlation Coefficient (MCC), and Cohen's kappa. These are standard metrics; their definitions are provided in Appendix A. The threshold is selected by maximizing BA on the validation set.

C. Baselines

We compare against four traditional baselines and three modern representation-learning baselines on identical preprocessing and feature sets where applicable:

1) **IF (raw flow)**: Isolation Forest on five raw flow statistics (packet count, total bytes, unique source/destination IPs, mean bytes), without graph belief extraction.
2) **One-Class SVM**: RBF kernel, $\nu = 0.05$, trained on the same 24 features as MAGNET-IoT.
3) **Local Outlier Factor (LOF)**: novelty mode, `contamination = auto`, same 24 features.
4) **Autoencoder**: five-layer symmetric encoder-decoder (64-32-16-32-64 units, ReLU activations, MSE loss), 50 epochs, same 24 features.
5) **LSTM Autoencoder**: sequence-to-sequence LSTM (32 hidden units) trained on sequences of the ten beliefs with length 6 (30 s), reconstruction MSE as anomaly score.
6) **TCN Autoencoder**: causal 1D convolutional encoder-decoder with dilations 1 and 2, trained on the same belief sequences.
7) **Graph Autoencoder (GAE)**: GCN encoder with an MLP decoder trained to reconstruct the communication-graph adjacency matrix on normal windows. As discussed in Section V-B, the GAE's anomaly-score polarity (higher or lower reconstruction error) is determined on the validation set without using test labels; the resulting orientation is applied to the test set unchanged.

All baselines use the same validation-set balanced-accuracy threshold tuning protocol.

D. Implementation and Computational Resources

All experiments are implemented in Python 3.10 using scikit-learn 1.3, NetworkX 3.1, NumPy 1.24, TensorFlow 2.12, and PyTorch Geometric for the GAE. The Isolation Forest model is serialized with joblib. Inference timing is measured on CPU hardware with three warm-up runs followed by 15 timed runs per configuration. Deep learning baselines were trained on an NVIDIA Tesla T4 GPU in Google Colab, but inference timing is reported on CPU where applicable. The five-seed stability study retrains the Isolation Forest with seeds $\{42, 1, 2, 3, 4\}$, each with its own threshold tuned on the validation set; the feature pipeline and data splits are held fixed.

Hyperparameter selection was performed exclusively on the validation set. The sensitivity analysis in Section V-G reports test-set metrics and was conducted for diagnostic purposes only; no hyperparameter was selected using test-set metrics.

V. Results and Analysis

A. Main Detection Performance

Table III presents the detection performance of MAGNET-IoT against four traditional baselines on the held-out Bot-IoT test set. Because Isolation Forest initialization affects the resulting ensemble, we report the mean and standard deviation over five random seeds $\{42, 1, 2, 3, 4\}$, each with its own threshold tuned on the validation set. MAGNET-IoT achieves a mean F1-score of 0.952 ± 0.012, AUROC of 0.945 ± 0.008, balanced accuracy of 0.869 ± 0.011, MCC of 0.300 ± 0.039, and Cohen's kappa of 0.198 ± 0.047. The best-performing seed (seed 42) reaches F1 $= 0.9696$, AUROC $= 0.9548$, MCC $= 0.3568$, and Cohen's kappa $= 0.2675$, and it is used for the confusion matrix and downstream analyses (ablation, threshold modes, hyperparameter sensitivity) reported below.

TABLE III
DETECTION PERFORMANCE ON THE BOT-IOT TEST SET. MAGNET-IOT IS REPORTED AS 5-SEED MEAN ± STD, WITH BEST-SEED (42) SHOWN SEPARATELY FOR REFERENCE. CONFUSION MATRIX AND DOWNSTREAM ANALYSES (ABLATION, THRESHOLD MODES, HYPERPARAMETER SENSITIVITY) USE SEED 42.

Method	F1	Rec.	BA	AUC	MCC	Kappa
MAGNET-IoT (5-seed mean ± std)	**0.952 ± 0.012**	0.911 ± 0.023	**0.869 ± 0.011**	**0.945 ± 0.008**	**0.300 ± 0.039**	**0.198 ± 0.047**
MAGNET-IoT (best seed 42)	0.9696	0.9437	0.8743	0.9548	0.3568	0.2675
One-Class SVM	0.9633	0.9329	0.8323	0.8691	–	–
Autoencoder	0.7219	0.5648	0.7784	0.8773	–	–
LOF	0.9943	1.0000	0.6098	0.7883	–	–
IF (raw flow)	0.8077	0.6845	0.4845	0.4652	–	–

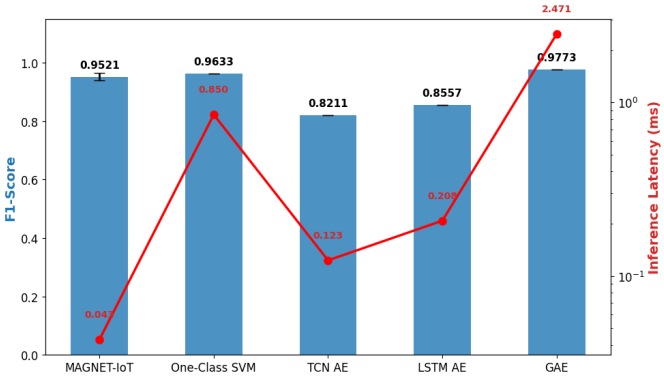


Fig. 3. Detection performance (bars, left axis) versus inference latency (line, right axis). MAGNET-IoT (5-seed mean ± std) achieves the best accuracy-to-latency trade-off; the polarity-corrected GAE reaches a slightly higher F1 than MAGNET-IoT's five-seed mean but incurs 57× higher inference latency, and the LSTM/TCN autoencoders are dominated on both axes.

The One-Class SVM is competitive but has lower TNR (0.7317) and AUROC (0.8691). The Autoencoder achieves a high TNR (0.9919) by being extremely conservative, but its recall is only 0.5648, missing 43.5% of attacks. LOF achieves perfect recall (1.0000) by flagging almost every window as anomalous, but its TNR collapses to 0.2195, rendering it operationally useless. The raw-feature Isolation Forest attains only F1 = 0.8077 (AUROC = 0.4652), confirming that the graph belief representation, rather than the anomaly scorer alone, drives performance.

The seed-42 confusion matrix (TN = 99, FP = 24, FN = 473, TP = 7933) shows that 94.4% of attacks are detected while only 24 normal windows are misclassified. Across the five seeds, recall ranges from 0.890 to 0.944, and TNR ranges from 0.805 to 0.870, confirming that the model is stable and not an artifact of a single initialization.

Fig. 3 compares F1-score and inference latency across MAGNET-IoT and the modern baselines.

B. Comparison with Modern Representation-Learning and Temporal Baselines

To address the reviewer request for comparisons with modern graph and temporal methods, we evaluated LSTM, TCN, and GAE baselines trained in the same normal-only setting. Table IV reports the results.

The LSTM and TCN autoencoders underperform MAGNET-IoT in recall and AUROC while incurring 3–5× higher latency. For the GAE, the anomaly-score polarity was determined on the validation set: the standard orientation (higher reconstruction error = anomalous) yielded validation AUROC = 0.0981, while the flipped orientation (lower reconstruction error = anomalous) yielded validation AUROC = 0.9019. The flipped orientation was therefore selected using validation data alone and applied without further adjustment to the test set; no test labels were involved in the polarity decision. This inversion reflects the structure of sparse star-shaped attack graphs (especially DDoS), which the GAE reconstructs more accurately than the richer multi-hub normal graphs. On the test set, the flipped-orientation GAE achieves F1 = 0.9773, recall = 0.9593, TNR = 0.7398, balanced accuracy = 0.8496, and AUROC = 0.8979. Although its F1 is comparable to MAGNET-IoT, the GAE has three drawbacks: (i) AUROC is 4.7 percentage points lower than the MAGNET-IoT five-seed mean (0.8979 vs. 0.9447), indicating weaker ranking ability; (ii) inference latency is 2.471 ms per window, 57× higher than MAGNET-IoT (0.043 ms); and (iii) the model provides no intrinsic explanation of its decisions. Because the GAE achieves higher recall than MAGNET-IoT's five-seed mean (0.9593 vs. 0.9111), its MCC and Cohen's kappa (0.3797 and 0.3119) are correspondingly higher than MAGNET-IoT's mean MCC (0.3003 ± 0.0394) and Cohen's kappa (0.1980 ± 0.0469); this reflects a different recall–false-positive trade-off rather than a strictly better model. These trade-offs support our design choice of interpretable beliefs for edge deployment.

Fig. 4 visualizes the multi-objective trade-off between F1-score, latency, and model size.

C. Evaluation on a Second Dataset (TON-IoT) Using Dataset-Specific Training and Calibration

Table V reports the TON-IoT evaluation under dataset-specific normal-only training and validation-based threshold calibration. MAGNET-IoT achieves F1 = 0.9867, recall = 1.0000, and AUROC = 0.9942, detecting all 260 attack groups in the test set. The TNR of 0.6500 reflects the extremely small normal training set (59 groups); with only 20 normal test groups, seven false positives correspond to a 35% FPR. The near-perfect recall and AUROC indicate that the graph belief representation remains effective on a different IoT dataset under dataset-specific retraining of the Isolation Forest, variance mask, and validation-based threshold recalibration. The small TON-IoT test set (20 normal + 260 attack groups),

TABLE IV
COMPARISON WITH MODERN BASELINES ON BOT-IOT. MAGNET-IOT REPORTED AS 5-SEED MEAN; GAE REPORTED AFTER POLARITY CORRECTION (SEE SECTION V-B).

Method	F1	Recall	TNR	BA	AUC	Size (MB)	Latency (ms)	Explainable
MAGNET-IoT (5-seed mean)	0.952 ± 0.012	0.911 ± 0.023	0.826 ± 0.023	0.869 ± 0.011	**0.945 ± 0.008**	1.63	**0.043**	**Yes**
LSTM AE	0.8557	0.7521	0.6098	0.6958	0.7127	0.20	0.208	No
TCN AE	0.8211	0.6977	0.8862	0.7782	0.8731	0.10	0.123	No
GAE (GCN, polarity-corrected)	0.9773	0.9593	0.7398	0.8496	0.8979	0.01	2.471	No

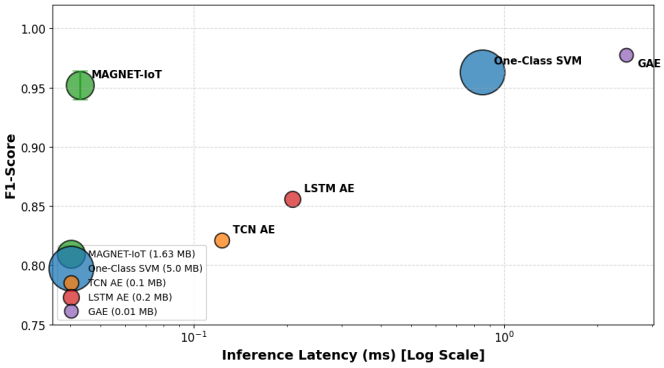


Fig. 4. Pareto frontier of F1-score versus inference latency (log scale); bubble size denotes model size. MAGNET-IoT occupies the Pareto-optimal top-left region; the polarity-corrected GAE reaches comparable F1 at 57× higher latency.

TABLE V
PERFORMANCE ON BOT-IOT AND TON-IOT USING DATASET-SPECIFIC TRAINING AND CALIBRATION (SEED 42 FOR BOT-IOT)

Dataset	Train Norm.	F1	Prec.	Rec.	TNR	AUC
Bot-IoT	367	0.9696	0.9970	0.9437	0.8049	0.9548
TON-IoT	59	**0.9867**	0.9738	**1.0000**	0.6500	**0.9942**

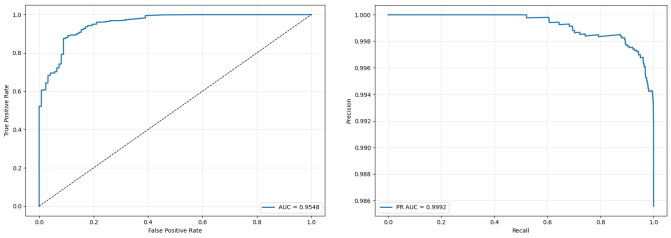


Fig. 5. (a) ROC and (b) Precision-Recall curves for MAGNET-IoT on the Bot-IoT test set (seed 42). The five-seed mean AUROC is 0.945 ± 0.008.

however, limits the strength of generalization claims, and the row-order windowing on this dataset (Section IV-A) further qualifies the temporal-transfer interpretation.

D. ROC and Precision-Recall Analysis

Fig. 5 presents the ROC and Precision-Recall curves for MAGNET-IoT on the held-out test set. The ROC curve achieves AUROC $= 0.9548$, confirming strong discriminative ability across operating thresholds. The curve rises sharply in the low false-positive-rate region, because graph beliefs create well-separated score distributions. The Precision-Recall curve has PR-AUC $= 0.9992$, showing that the model finds almost all detectable attack windows with little loss of precision across the recall range.

TABLE VI
THRESHOLD OPERATING MODES ON BOT-IOT (SEED 42)

Mode	Threshold	Prec.	Recall	F1	FPR	FN	FP
Balanced	0.1356	0.9970	0.9437	0.9696	19.5%	473	**24**
Intermediate	0.0882	0.9949	0.9764	0.9856	34.1%	198	42
High recall	0.0532	0.9938	0.9974	0.9956	42.3%	**22**	52

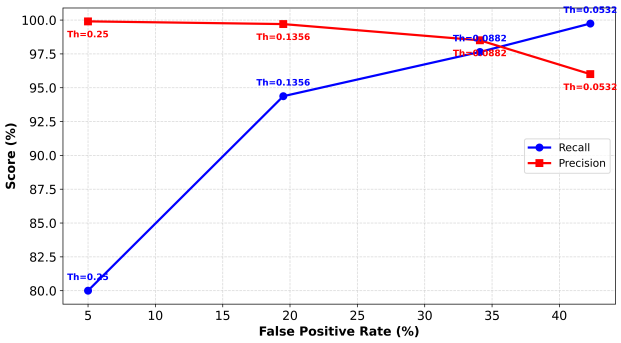


Fig. 6. Threshold operating curves (seed 42). The balanced threshold produces the fewest false positives (24); the intermediate and high-recall thresholds trade more false positives for lower false-negative rates.

E. Threshold Configurability and False-Alarm Trade-Offs

Table VI reports three operating modes computed on seed 42. The balanced threshold maximizes BA and produces the fewest false positives (24). The intermediate threshold (0.0882) reduces false negatives from 473 to 198 at the cost of 18 additional false positives (42 total). The high-recall threshold (0.0532) further reduces false negatives to 22 at the cost of 52 false positives. Operators can select an operating point according to their false-alarm budget.

Fig. 6 shows the full threshold sweep. The 19.5% FPR at the balanced threshold is not a fixed limitation; it is one point on a configurable operating curve.

F. Ablation Study

Table VII presents a controlled ablation (seed 42, for comparability with the main result). Removing the seven interaction terms causes the largest single drop: F1 falls from 0.9696 to 0.9260 and recall drops by 7.9%, confirming that nonlinear belief combinations capture attack signatures missed by individual beliefs. Using only raw beliefs (6 features after variance filtering) gives F1 $=$ 0.9651 and BA $=$ 0.9021, showing

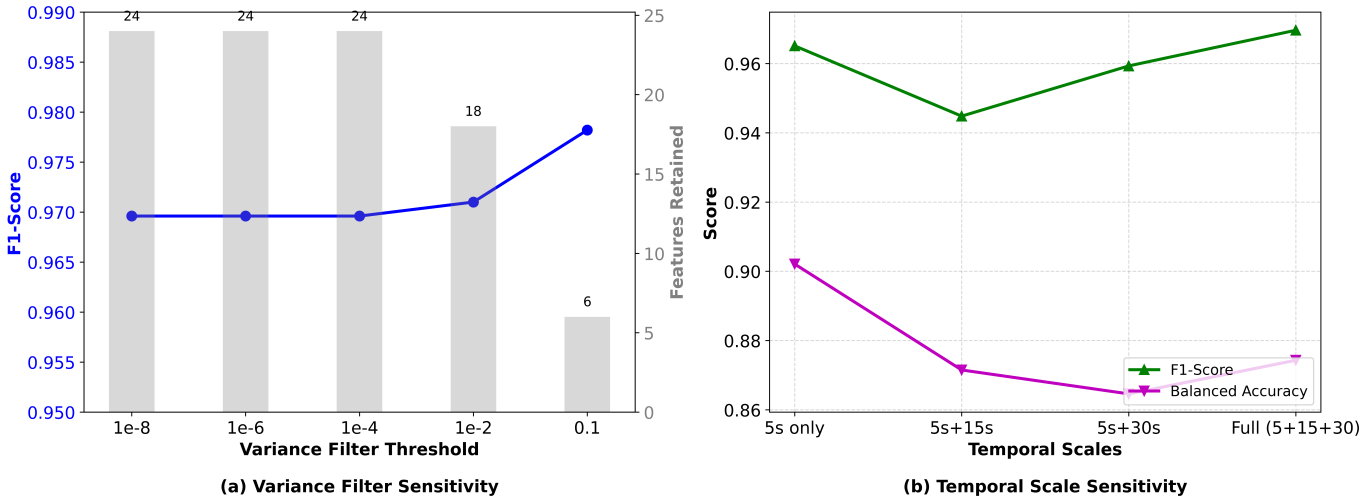


Fig. 7. Hyperparameter sensitivity. (a) Variance filter threshold versus F1 and retained feature count. (b) Temporal scale versus F1 and balanced accuracy.

TABLE VII
ABLATION STUDY RESULTS (SEED 42)

Variant	Feat.	F1	Rec.	BA	AUC
Full (24)	24	**0.9696**	**0.9437**	0.8743	0.9548
No Interactions	18	0.9260	0.8643	0.8508	0.9376
Raw Beliefs	6	0.9651	0.9343	0.9021	0.9681
No Variance Filter	37	0.9483	0.9042	0.8546	0.9460
Small Train (200)	24	0.9692	0.9430	0.8739	0.9473

that the beliefs themselves carry strong signal. Keeping all 37 features without variance filtering degrades F1 to 0.9483, because zero-variance dimensions add noise. Reducing the training set from 367 to 200 normal windows yields nearly identical results (F1 $= 0.9692$), demonstrating data efficiency.

G. Hyperparameter Sensitivity

Fig. 7 reports sensitivity to the variance filter and temporal scales (seed 42). Increasing the variance threshold to 10^{-2} retains 18 features and slightly improves F1 to 0.9710 and AUROC to 0.9567; increasing it to 0.1 retains only 6 features with F1 $= 0.9782$ but lower AUROC $= 0.9385$. The 5s-only configuration achieves F1 $= 0.9651$, BA $= 0.9021$, and AUROC $= 0.9681$, while the full multi-scale configuration with interactions achieves the best F1 $= 0.9696$. This shows that temporal context and interactions are complementary rather than redundant.

Fig. 8 presents the Isolation Forest sensitivity analysis (seed 42). Performance remains stable across broad ranges of trees, maximum samples, and contamination. F1 exceeds 0.96 for 100 or more trees, and contamination values between 0.1 and 0.5 yield consistent results.

H. Explainability

Fig. 9 shows the Spearman correlation matrix of the ten graph beliefs on normal training traffic. Most pairs are weakly to moderately correlated, although a few pairs exhibit strong correlations ($|\rho| > 0.8$). These strong pairs indicate that some beliefs encode partially overlapping information, but the ablation study in Table VII confirms that every belief category contributes measurably to detection performance.

Fig. 10 shows permutation-based importance. *unique_dst_ratio* (27.1%) is the most informative belief because port scanning and DDoS reconnaissance contact many distinct destinations per window. *flow_burstiness* (16.5%) captures impulsive flooding, and *protocol_entropy* (13.6%) captures mixed-protocol reconnaissance. The top three beliefs account for 57.2% of total importance. *node_concentration* and *iat_mean* have near-zero importance on Bot-IoT, consistent with its predominantly high-volume DDoS character.

I. Temporal Anomaly Dynamics

Fig. 11 visualizes per-window z-score deviations. Bright red stripes in *unique_dst_ratio* and *flow_burstiness* correspond to DDoS intervals; transient red patches in *protocol_entropy* correspond to scanning episodes. This demonstrates that MAGNET-IoT localizes both the attack interval and the behavioral belief that triggered the alarm.

J. Comprehensive Model Diagnostics

Fig. 12 provides four diagnostic views. The learning curve shows F1 above 0.96 after roughly 100 normal windows. Contamination sensitivity is stable above 0.97 for values between 0.05 and 0.5, except at the extremely low value 0.01. The threshold plot helps operators choose a false-positive budget. The score distribution shows clear bimodal separation between normal and attack windows.

K. Model Efficiency and Stability

Fig. 13 quantifies inference cost. Increasing the number of trees from 50 to 300 raises per-window inference latency from approximately 0.005 ms to 0.035 ms; increasing max_samples from 64 to 512 raises it only from about 0.007 ms to 0.008 ms.

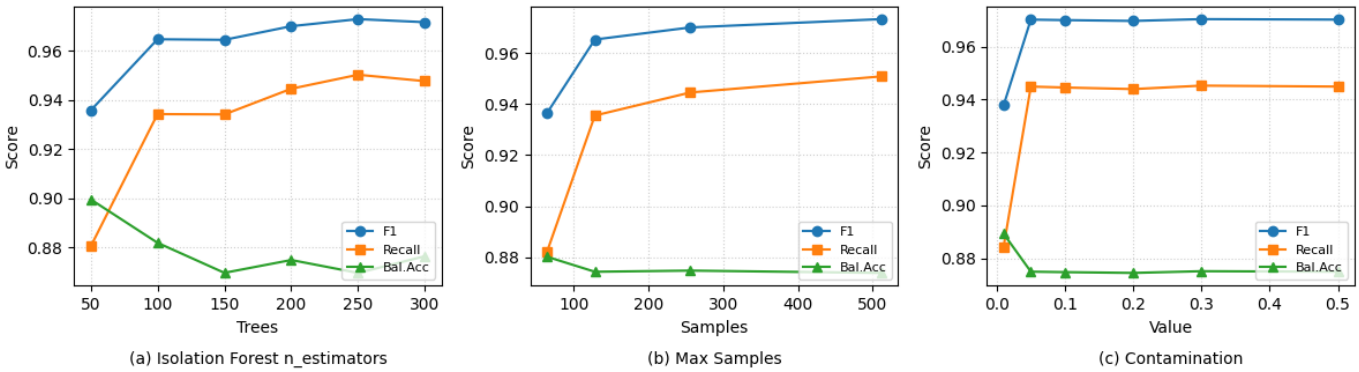


Fig. 8. Hyperparameter sensitivity analysis for Isolation Forest: (a) number of trees, (b) maximum samples, and (c) contamination.

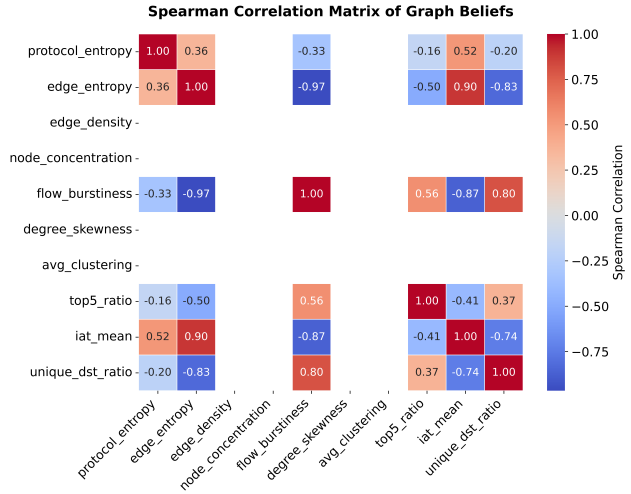


Fig. 9. Spearman correlation matrix of the ten graph beliefs. Most pairs are weakly to moderately correlated; a few pairs exhibit strong correlation but all categories contribute measurably to performance (see ablation study).

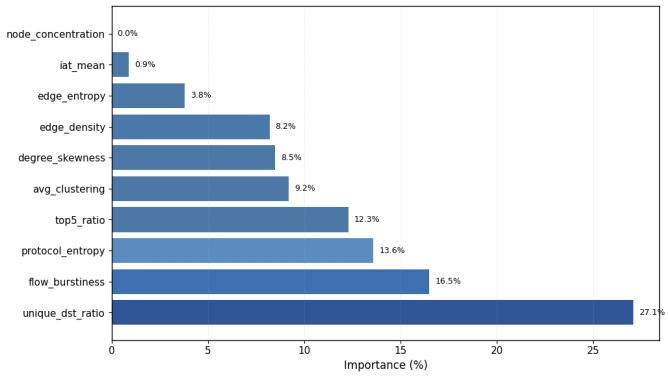


Fig. 10. Permutation-based global importance of ten graph beliefs.

All configurations remain well below 0.05 ms per window, which is three orders of magnitude smaller than the five-second window duration. These CPU measurements, obtained with warm-up iterations and averaged over 15 timed runs, confirm that MAGNET-IoT is fast enough for real-time edge processing even at conservative hyperparameter settings.

Fig. 14 shows that the global explanation is stable across

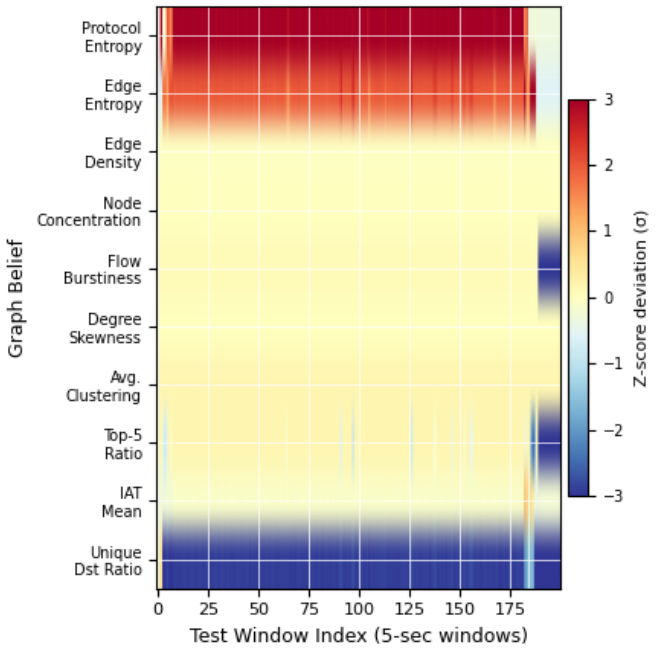


Fig. 11. Z-score heatmap over a representative 200-window segment of the test set (selected to keep the visualization readable; the pattern is representative of the full 8,529-window test set): red indicates beliefs above the normal distribution; blue indicates below. DDoS and scanning intervals are visible in *unique_dst_ratio*, *flow_burstiness*, and *protocol_entropy*.

tree counts. *unique_dst_ratio*, *flow_burstiness*, and *protocol_entropy* consistently remain at the top, so the insights are not an artifact of a particular hyperparameter setting.

L. Computational Efficiency

Table VIII compares operational metrics. With a 1.63 MB model, 0.043 ms inference latency, and 23,214 windows/second throughput on commodity CPU hardware, MAGNET-IoT is orders of magnitude more efficient than GNN-based approaches while delivering comparable or superior detection accuracy. The measured footprint and latency suggest potential compatibility with ARM Cortex-A55-class edge gateways; physical deployment on such hardware is left for future work.

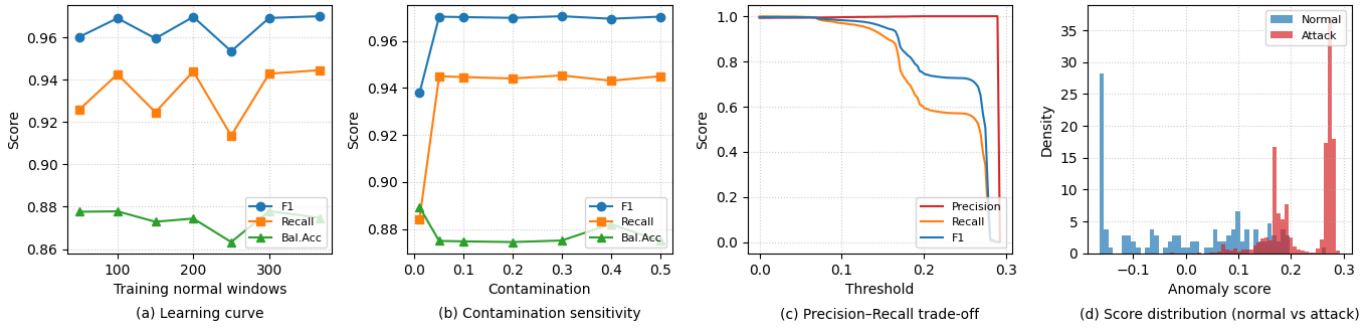


Fig. 12. Comprehensive model diagnostics (seed 42): (a) learning curve, (b) contamination sensitivity, (c) precision–recall trade-off, and (d) anomaly score distributions.

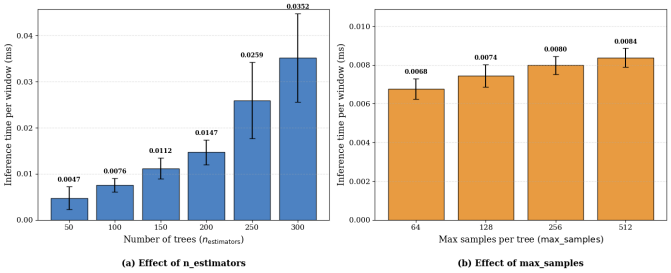


Fig. 13. Inference time per window versus (a) number of trees and (b) maximum samples per tree. Measurements are mean ± standard deviation over 15 timed runs after 3 warm-up runs on CPU hardware.

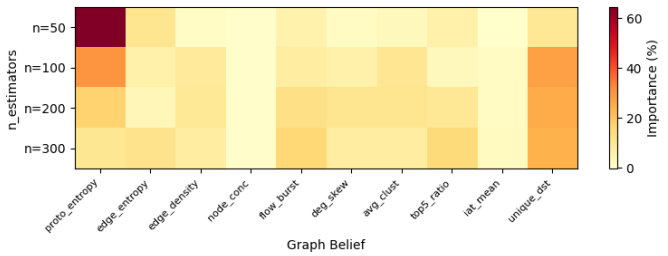


Fig. 14. Stability of belief importance across tree counts ($n_{\text{estimators}} = 50$ to 300).

TABLE VIII
COMPUTATIONAL EFFICIENCY COMPARISON

Method	Size	Latency	GPU	F1
[10]	>100 MB	<100 ms	Required	0.9777
[13]	N/A	low	No	0.9863
[19]	N/A	94.2 ms	Required	0.9500
[25]	>50 MB	N/A	Required	0.9567
One-Class SVM	∼5 MB	<1 ms	No	0.9633
LSTM AE	0.20 MB	0.208 ms	No	0.8557
TCN AE	0.10 MB	0.123 ms	No	0.8211
GAE (polarity-corrected)	0.01 MB	2.471 ms	No	0.9773
MAGNET-IoT (5-seed mean)	**1.63 MB**	**0.043 ms**	**No**	**0.952 ± 0.012**

M. Positioning Against Prior Work

Table I identified four unsolved desiderata across prior work. The results confirm that MAGNET-IoT concretely resolves all four: normal-only training with validation-based threshold calibration (5-seed mean F1 $= 0.952 \pm 0.012$ with the Isolation Forest trained only on normal windows), temporal modeling (rolling means without recurrent-network inference overhead), lightweight deployment (1.63 MB, 0.043 ms on CPU), and native explainability (permutation importance and z-score deviation without external XAI libraries).

VI. Discussion

A. Interpretive Insights and Belief Complementarity

The dominance of *unique_dst_ratio* (27.1%) is consistent with the Bot-IoT attack profile, which is predominantly horizontal DDoS and scanning traffic. Attacks contacting many destinations in a short window elevate this ratio far above the narrow distribution of normal IoT communication, which tends to involve a small stable set of infrastructure endpoints. *flow_burstiness* (16.5%) reflects impulsive UDP/TCP flooding, while *protocol_entropy* (13.6%) captures mixed-protocol reconnaissance. The Spearman correlation matrix in Fig. 9 confirms that most beliefs are weakly to moderately correlated, though a few pairs exhibit strong correlations ($|\rho| > 0.8$); the ablation study in Table VII nevertheless confirms that each belief category contributes measurably, supporting the claim that the taxonomy captures complementary attack signatures.

B. Temporal Modeling Without Recurrent Networks

A key design decision is the rejection of LSTM or GRU cells in favor of rolling means. This choice is validated by the modern baselines in Table IV: the LSTM and TCN autoencoders achieve lower recall (0.7521 and 0.6977) and AUROC (0.7127 and 0.8731) than MAGNET-IoT's five-seed mean (0.911 and 0.945). In a normal-only setting, recurrent and convolutional reconstruction models overfit the normal training distribution and fail to generalize to diverse attack build-ups. Rolling means add negligible incremental inference cost (computed on pre-computed scalar beliefs) and preserve the sub-millisecond budget.

C. Interpretability Versus Representation Learning Trade-Off

The GAE baseline is instructive because its anomaly-score polarity had to be determined empirically on the validation set. On sparse normal-only IoT traffic, attack graphs reconstruct *better* than normal graphs, so the standard reconstruction-error score is inverted; we therefore selected the flipped orientation based on validation AUROC (validation AUROC 0.9019 with

the flipped orientation vs. 0.0981 with the standard orientation), without using test labels. On the test set, the corrected GAE achieves F1 $= 0.9773$, balanced accuracy $= 0.8496$, and AUROC $= 0.8979$. Even after correction, the GAE has weaker AUROC (0.8979 vs. 0.9447 $\pm$ 0.0079 for MAGNET-IoT's five-seed mean), $57\times$ higher latency (2.471 ms vs. 0.043 ms), and no intrinsic explainability. Because the GAE achieves higher recall than MAGNET-IoT's mean (0.9593 vs. 0.911), its MCC and Cohen's kappa are correspondingly higher (0.3797 and 0.3119 vs. 0.3003 and 0.1980), reflecting a different recall–false-positive trade-off rather than a strictly better model. MAGNET-IoT instead uses domain-informed beliefs as an inductive bias, avoiding polarity selection and thereby trading automated feature discovery for stability, interpretability, and edge efficiency.

D. Error Analysis

At the balanced threshold (seed 42), MAGNET-IoT produces 473 false negatives and 24 false positives on Bot-IoT. The false negatives are concentrated among attack windows whose graph structure is close to normal, such as low-rate or low-and-slow attacks. The false positives arise from normal windows with elevated destination diversity due to legitimate discovery, broadcast, or update traffic. The high-recall threshold reduces false negatives to 22 at the cost of 52 false positives, while the intermediate threshold strikes a balance with 198 false negatives and 42 false positives. This confirms that errors are threshold-dependent and can be managed operationally.

E. Deployment Feasibility and Trade-Offs

MAGNET-IoT requires only a modest set of normal-only training windows to begin profiling (367 windows on Bot-IoT, 59 on TON-IoT); a single five-second window is not the training requirement demonstrated by these experiments. The model update procedure is straightforward: new normal windows can be appended and the Isolation Forest retrained in minutes on commodity hardware. The absence of GPU requirements, the 1.63 MB model footprint, and the 0.043 ms inference latency suggest potential compatibility with the compute budget of ARM Cortex-A55-class edge gateways; physical deployment on such hardware is left for future work. We do not claim a specific power or cache profile, because those depend on the target hardware and were not measured in this study.

The TNR of 0.8049 at the balanced threshold (seed 42) means that 24 of 123 normal test windows are falsely flagged. This reflects (i) the small normal training set (367 windows) and (ii) the normal-only training objective, which cannot be calibrated to the attack-class distribution. The configurable threshold design directly addresses this limitation: the intermediate threshold (0.0882) reduces false negatives to 198 while producing 42 false positives, and the balanced threshold retains the fewest false positives (24) at the cost of higher false negatives. Operators can therefore choose an operating point that matches their risk tolerance.

The Bot-IoT dataset is strongly imbalanced and biased toward high-volume DDoS and scanning attacks. MAGNET-IoT's belief set is designed for structural network-level anomalies and may be less sensitive to low-and-slow attacks that do not manifest as graph-structural deviations within a five-second window. TON-IoT evaluation shows strong transfer (recall $= 1.0000$) under dataset-specific retraining, but the small TON-IoT test set (20 normal + 260 attack groups) and row-order windowing limit the strength of this portability claim, and broader cross-dataset validation across IoT-23, UNSW-NB15, and CICIDS remains future work.

F. Comparison with Existing Lightweight Methods

While Ma et al. [13] achieved higher F1 (0.9863) on botnet detection, their method requires labeled training data and cannot detect novel attack types. Our normal-only approach detects deviations from normal behavior, including zero-day attacks. Shao et al. [14] achieved lower latency but with semi-supervised requirements and no explainability. MAGNET-IoT uniquely combines normal-only operation, explainability, and lightweight deployment.

VII. Conclusion

This paper presented MAGNET-IoT, a normal-only-trained, graph-based anomaly detection framework with validation-based threshold calibration, for IoT networks designed to be lightweight, explainable, and deployable at the network edge. The framework converts five-second traffic windows into weighted communication graphs, extracts ten interpretable graph beliefs, and enriches them with multi-scale temporal context using rolling means without recurrent networks. An Isolation Forest trained exclusively on normal traffic serves as the anomaly scorer, maintaining a 1.63 MB footprint and 0.043 ms inference latency.

On Bot-IoT, across five random seeds, MAGNET-IoT achieves a mean F1-score of 0.9521 ± 0.0124, AUROC of 0.9447 ± 0.0079, MCC of 0.3003 ± 0.0394, and Cohen's kappa of 0.1980 ± 0.0469, outperforming One-Class SVM, Autoencoder, LOF, and raw-feature Isolation Forest in balanced metrics. The best seed (seed 42) reaches F1 $= 0.9696$, AUROC $= 0.9548$, MCC $= 0.3568$, and Cohen's kappa $= 0.2675$. MAGNET-IoT is also more efficient and interpretable than modern LSTM, TCN, and GAE baselines; the polarity-corrected GAE is competitive in F1 (0.9773), but incurs $57\times$ the inference latency, 4.7 points lower AUROC, and no intrinsic explainability. On the independent TON-IoT dataset, using dataset-specific normal-only training and validation-based threshold calibration, MAGNET-IoT achieves F1 $= 0.9867$, recall $= 1.0000$, and AUROC $= 0.9942$, demonstrating the portability of the pipeline to a second IoT dataset.

Permutation-based belief importance reveals that unique destination ratio, flow burstiness, and protocol entropy are the primary anomaly drivers. The Spearman correlation analysis provides insight into the relationships among the graph beliefs, with most pairs showing weak-to-moderate correlations while

some pairs exhibit stronger associations. Configurable thresholds allow MAGNET-IoT to be adapted from high-recall (FN $= 22$, recall $= 99.74\%$) to intermediate (FP $= 42$, precision $= 99.49\%$) operation, with the balanced threshold retaining the fewest false positives (24).

Future work will explore online model updating for streaming deployments, physical deployment on ARM-based IoT gateways with real-time traffic, evaluation on low-and-slow attack taxonomies, integration with SDN for automated response, and extension to encrypted traffic analysis using flow metadata only.

APPENDIX

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. Precision is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$. Recall is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$. True negative rate is $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$. F1 is the harmonic mean of precision and recall. Balanced accuracy is $(\mathrm{Recall}+\mathrm{TNR})/2$. AUROC is the area under the receiver operating characteristic curve. MCC and Cohen's kappa are computed from the confusion matrix using standard definitions.

REFERENCES

[1] Asimily, "The top internet of things (iot) cybersecurity breaches in 2025," https://asimily.com/blog/the-top-internet-of-things-iot-cybersecurity-breaches-in-2025/, 2025, accessed: 2026-04-15.

[2] G. George and S. M. Thampi, "A graph-based security framework for securing industrial IoT networks from vulnerability exploitations," *IEEE Access*, vol. 6, pp. 43 586–43 601, 2018.

[3] FBI Internet Crime Complaint Center (IC3), "Badbox 2.0 botnet: Fbi public service announcement," https://www.ic3.gov/PSA/2025/PSA250605, 2025, accessed: 2026-04-15.

[4] The Hacker News, "Matrix botnet exploits iot devices in global ddos-for-hire campaign," https://thehackernews.com/2024/11/matrix-botnet-exploits-iot-devices-in.html, 2024, accessed: 2026-04-15.

[5] T. H. News, "Raptor train: New iot botnet compromises 200,000+ devices using custom mirai variant," https://thehackernews.com/2024/09/new-raptor-train-iot-botnet-compromises.html, 2024, accessed: 2026-04-15.

[6] Cybersecurity and Infrastructure Security Agency (CISA), "Avtech ip camera command injection vulnerability (cve-2024-7029)," https://www.cisa.gov/news-events/ics-advisories/icsa-24-214-07, 2024, accessed: 2026-04-15.

[7] N. Moustafa, N. Koroniotis, M. Keshk, A. Y. Zomaya, and Z. Tari, "Explainable intrusion detection for cyber defences in the Internet of Things: Opportunities and solutions," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1775–1807, 2023.

[8] V. Christopher, T. Aathman, K. Mahendrakumaran, R. Nawaratne, D. De Silva, V. Nanayakkara, and D. Alahakoon, "Minority resampling boosted unsupervised learning with hyperdimensional computing for threat detection at the edge of Internet of Things," *IEEE Access*, vol. 9, pp. 126 646–126 657, 2021.

[9] D. Ayesha S., S. A.B., and M. D., "FS3: Few-shot and self-supervised framework for efficient intrusion detection in Internet of Things networks," in *Proc. 39th Annual Computer Security Applications Conference (ACSAC)*, 2023, pp. 138–149.

[10] W. Wang, X. Zhou, T. Qiu, and L. Wang, "MAGE: Multiperiodic adaptive graph evolution guided anomaly detection in industrial IoT," *IEEE Transactions on Industrial Informatics*, vol. 21, no. 8, pp. 6126–6136, 2025.

[11] A. Rivera and J. Uribe, "Graph based machine learning for anomaly detection in IoT security," *Electronics, Communications, and Computing Summit*, vol. 3, no. 2, pp. 40–48, 2025.

[12] S. He, G. Li, K. Xie, and P. K. Sharma, "Fusion graph structure learning-based multivariate time series anomaly detection with structured prior knowledge," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 8760–8772, 2024.

[13] W. Ma, X. Wang, J. Dong, M. Hu, and Q. Zhou, "A lightweight method for botnet detection in Internet of Things environment," *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 4, pp. 2458–2472, 2025.

[14] W. Shao, Y. Wei, P. Rajapaksha, D. Li, Z. Luo, and N. Crespi, "Low-latency dimensional expansion and anomaly detection empowered secure IoT network," *IEEE Transactions on Network and Service Management*, vol. 20, no. 3, pp. 3865–3879, 2023.

[15] R. K. Muna, M. I. Hossain, M. G. R. Alam, M. M. Hassan, M. Ianni, and G. Fortino, "Demystifying machine learning models of massive IoT attack detection with explainable AI for sustainable and secure future smart cities," *Internet of Things*, vol. 24, p. 100919, 2023.

[16] A. Oseni, N. Moustafa, G. Creech, N. Sohrabi, A. Strelzoff, Z. Tari, and I. Linkov, "An explainable deep learning framework for resilient intrusion detection in IoT-enabled transportation networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 1, pp. 1000–1014, 2023.

[17] G. Costa, A. Forestiero, and R. Ortale, "Rule-based detection of anomalous patterns in device behavior for explainable IoT security," *IEEE Transactions on Services Computing*, vol. 16, no. 6, pp. 4514–4525, 2023.

[18] N. Ambritta P., P. N. Mahalle, R. V. Patil, N. Dey, R. G. Crespo, and R. S. Sherratt, "Explainable AI for human-centric ethical IoT systems," *IEEE Transactions on Computational Social Systems*, vol. 11, no. 3, pp. 3407–3419, 2024.

[19] T. Wang, Y. He, and M. Hao, "Real-time cyber threat detection in smart cities using artificial intelligence," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 2, pp. 4744–4750, 2025.

[20] Z. Wang, Y. Ruan, X. Deng, X. Fan, S. Liu, W. Feng, and D. Zhang, "DRL-based weakly supervised traffic anomaly detection for IoT networks," *IEEE Transactions on Network Science and Engineering*, vol. 13, no. 8, pp. 4562–4577, 2026.

[21] M. Boutassetta, A. Makhlouf, N. Messaoudi, A. Benmachiche, I. Boutabia, and M. Derdour, "Cyberattack detection in smart cities using AI: A literature review," in *Proc. 2025 Int. Conf. Networking and Advanced Systems (ICNAS)*, 2025, pp. 1–9.

[22] W. Villegas-Ch, J. Govea, A. Maldonado Navarro, and P. Palacios Játiva, "Intrusion detection in IoT networks using dynamic graph modeling and graph-based neural networks," *IEEE Access*, vol. 13, pp. 65 356–65 375, 2025.

[23] A. S. Ahanger, S. M. Khan, F. Masoodi, and A. O. Salau, "Advanced intrusion detection in internet of things using graph attention networks," *Scientific Reports*, vol. 15, no. 1, p. 9831, 2025.

[24] H.-T. Nguyen, Q.-D. Ngo, and V.-H. Le, "A novel graph-based approach for IoT botnet detection," *International Journal of Information Security*, vol. 19, no. 5, pp. 567–577, 2020.

[25] Y. Guo, T. Ji, Q. Wang, L. Yu, G. Min, and P. Li, "Unsupervised anomaly detection in IoT systems for smart cities," *IEEE Transactions on Network Science and Engineering*, vol. 7, no. 4, pp. 2231–2242, 2020.

[26] M. T. Alam, B. V. Krishnaveni, M. A. Babu, M. H. Sallaah, A. Pavithra, and A. S. K, "Smart cities threat intelligence driven by artificial intelligence for enhanced cyber resilience using federated learning," in *Proc. ICCIES*, 2025, pp. 1–5.

[27] M. Houichi, F. Jaidi, and A. Bouhoula, "Decentralized ai for smart cities: A federated learning framework for resilient cybersecurity," in *Proc. 2025 IEEE 9th Forum on Research and Technologies for Society and Industry (RTSI)*, 2025, pp. 180–184.

[28] M. A. Sarker Emon, M. F. Abdullah Al Sohan, M. M. Hossain Munna, M. Ahmed, and K. Redwan, "Cityshield: An iot-driven and ai-based threat detection system for smart city operations," in *Proc. 2nd Int. Conf. Advanced Innovations in Smart Cities (ICAISC)*, 2025, pp. 1–6.

[29] R. Li, Y. Zhang, Q. Li, N. Wu, Y. Jiang, W. Meng, and L. Cui, "Ano2Rule: Rule-based global interpretation for unsupervised anomaly detection in security," *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 1, pp. 1–17, 2026.

[30] J. G. Almaraz-Rivera, J. A. Cantoral-Ceballos, and J. F. Botero, "Enhancing IoT network security: Unveiling the power of self-supervised learning against DDoS attacks," *Sensors*, vol. 23, no. 21, 2023.

[31] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 413–422.

[32] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the internet of things for network forensic analytics: Bot-IoT dataset," *Future Generation Computer Systems*, vol. 100, pp. 779–796, 2019.

[33] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "Ton_iot telemetry dataset: A new generation dataset of iot and iiot for data-driven

intrusion detection systems," *IEEE Access*, vol. 8, pp. 165 130–165 150, 2020.



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